

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE

CONTINUOUS ASSESSMENT TEST - I - MAY-2026

CET 05201 - COMPUTER NETWORKS

SEMESTER: II

DURATION: 1 Hr 30 Min.

YEAR: II

MAX. MARKS: 50

PART - A (Answer all the Questions) (7 x 2 = 14)

1. Find the number of connections needed for a full mesh topology with 6 computers.
2. Explain the purpose of Network Address Translation (NAT) in networking?
3. In your own words, explain why IP address division into smaller networks is important in large organizations? Two (2) reasons.
4. Examine a function of Packet InterNet Groper.
5. Compare Unshielded Twisted Pair (UTP) and Shielded Twisted Pair (STP) cables in terms of structure and interference protection.
6. What does the abbreviation RJ45 stand for in networking?
7. For the IP 192.168.50.10/27, calculate the network ID.

PART - B (Answer any three Questions) (3 x 4 = 12)

8. A technician wants to test whether a network cable is properly connected before establishing a

LAN network. Identify a suitable diagnostic equipment and explain its function.

9. Explore the key factors to be considered in the design of transmission media.

10. Re-write all the Private IP address from the list.

- I. 172.16.4.0 IV. 172.32.19.4
- II. 10.4.16.32 V. 192.169.0.0
- III. 192.168.100.0 VI. 172.31.15.6

11. Compare guided media and unguided media by giving four (4) differences.

12. Differentiate between Simplex, Half Duplex, and Full Duplex communication modes, giving one example for each.

13. Given the IP address 192.168.2.9 with subnet mask 255.255.255.252.

- A) Determine the network address
- B) Determine the broadcast address
- C) State the number of usable hosts in the subnet